

QUANTITATIVE MICROBIAL ANALYSIS AND BACTERIAL CHARACTERIZATION OF LONG-SLEEVED LAB GOWNS AFTER VARIED DURATIONS OF USE IN MIAGAO, ILOILO, PHILIPPINES

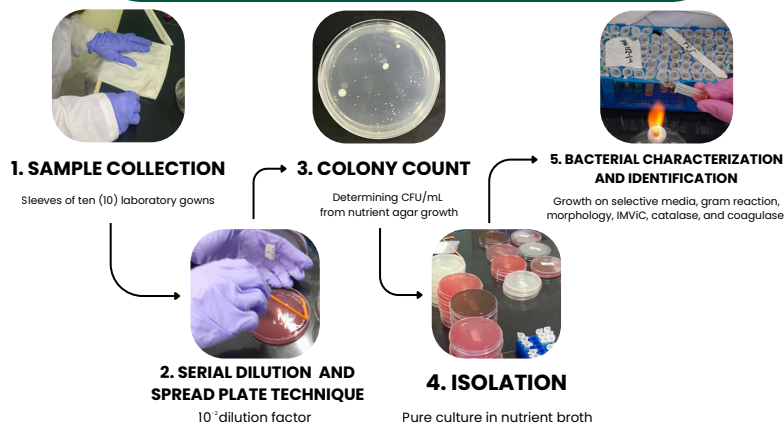
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INTRODUCTION

Laboratory gowns are **essential for safety in academic and professional settings**, protecting against hazardous agents like biological specimens and chemicals. However, they can harbor microorganisms, **potentially spreading microbes if not properly maintained**.

This study investigates **microbial contamination on lab gown cuffs**, focusing on how *usage duration* and *washing methods* (*handwashing* vs. *machine washing*) affect **bacterial loads**. The results aim to **improve hygiene practices** in academic laboratories and **minimize contamination risks**.

MATERIALS AND METHODS



RESULTS AND DISCUSSION

The results show that **bacterial contamination stabilizes or decreases after extended use**, possibly due to *microbial competition* and *environmental exposure*.

Machine washing at lower temperatures may result in **biofilm persistence**, contributing to **higher contamination levels**.

Predominant isolates included *S. aureus*, *S. epidermidis*, and *Bacillus spp.*, accounting for **62.5% of all isolates**. These species represent both *human-associated microbiota* and *environmental contaminants*.

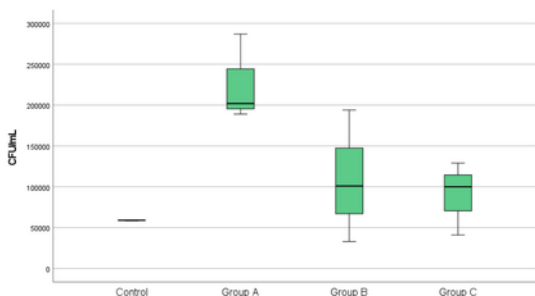


Figure 1. Mean Bacterial Load (CFU/mL) Across Groups.

Bacterial load, expressed as CFU/mL, was **highest in Group A (1–2 weeks of use)** with a mean of $226,000 \pm 53,226$ CFU/mL. This was **significantly higher than the Control** (freshly washed gown, 59,000 CFU/mL; $p = 0.021$). **Groups B (3–4 weeks)** and **C (4–6 weeks)** showed **lower contamination levels**, with **no significant differences** compared to the **Control**.

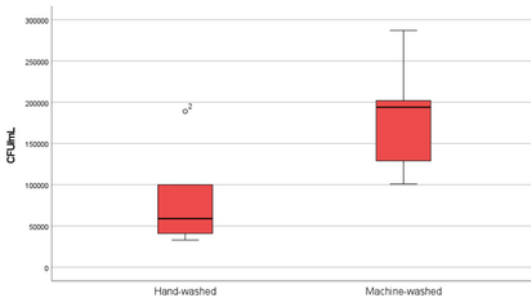
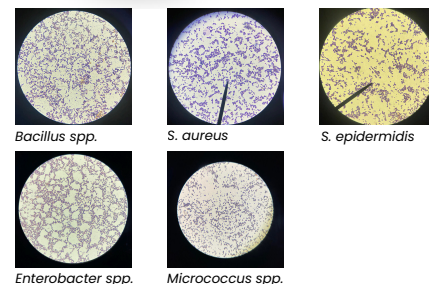
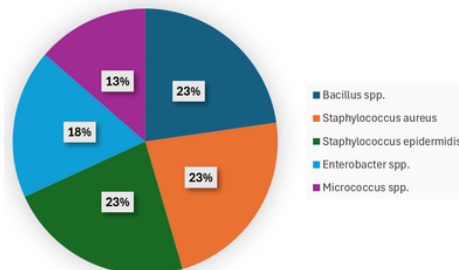


Figure 2. Mean Bacterial Load (CFU/mL) by Washing Method.

Bacterial load, expressed as CFU/mL, was **highest in Group A (1–2 weeks of use)** with a mean of $226,000 \pm 53,226$ CFU/mL. This was **significantly higher than the Control** (freshly washed gown, 59,000 CFU/mL; $p = 0.021$). **Groups B (3–4 weeks)** and **C (4–6 weeks)** showed **lower contamination levels**, with **no significant differences** compared to the **Control**.



CONCLUSION

- Machine washing** generally shows **higher contamination than handwashing**, likely due to *varying washing conditions* (e.g., temperature, detergent).
- Predominant bacteria include *Staphylococcus aureus*, *Staphylococcus epidermidis*, and *Bacillus spp.*, reflecting contamination from **human contact and the environment**.
- Proper washing protocols**, including **high temperatures (>60°C)**, are **essential to minimize microbial loads**.
- Enhanced hygiene practices** and **educational initiatives** can **improve laboratory safety** and **reduce cross-contamination risks**.

RECOMMENDATIONS

- Standardize washing protocols** (temperature, detergent type, drying methods).
- Use **molecular methods** (e.g., 16S rRNA sequencing) for **precise bacterial identification**.
- Include **larger sample sizes** for future studies to **enhance reliability**.

REFERENCES

